Southern Cross University

Workbook on multi-part tutorial questions

You must read the full assessment brief first. The following information is only a reminder of the most important requirements and does not replace the brief.

- **Location of the Questions**
  - **The assessment questions are only available in:**
    - **The live tutorial, and**
    - **The video tutorial series on Blackboard.**
  - **The questions are not posted anywhere else.**
  - **The questions are not available until the tutorial has been delivered.**

- **Which Questions Must be Answered**
  - **You must answer Multi-part Tutorial Question B.**
  - **You must also choose one of the remaining two questions:**
    - **Question A *or* Question C.**

This means you will answer two multi-part questions:

- **A & B, or**
- **B & C**

For each multi-part question you choose, you must answer every sub-question within it.

- **Word Limits**
  - **600 words total inclusive of all sub-questions in Question A *or* C, <u>and</u>**
  - **600 words total inclusive of all sub-questions in Question B.**
    - **For a total of 1200 words in your workbook**

The distribution of these words for each sub-question is up to you. Do not include the question text in the word count.

- **How to Enter Your Answers**
  - **Type the full question next to the appropriate sub-question number.**
  - **Enter only your answer in the answer box provided.**
  - **The first few words of each question are pre-filled in the workbook to help ensure you are answering the correct question.**

Example format:

1. Using the antihistamines discussed in the tutorial, compare…

(continue to type in the remaining portion of the question, and then type your answer in the answer box)

Type your answer in the box.

- **Writing Requirements**
  - **Do NOT use point form.**
    - **Point form will receive ZERO marks.**
      - **You do not speak to a client in point form, and your writing should reflect this.**
  - **Answers must be written in full sentences and paragraphs.**

- **Content Expectations**
  - **Reflect carefully on safety:**
    - **Are there important safety issues associated with any medication?**
    - **For instance, is there a risk of serious health implications with regular use?**
    - **The safety issues are highly weighted and should be clearly emphasised in your answers.**
  - **If a question asks for the "most important side effects", include:**
    - **The most serious, and**
    - **The most commonly experienced side effects.**

- **Use of Tutorial Content and References**

- **Your answer must emphasise the key points highlighted in Tutorial Week 3 (Question A), Tutorial Week 4 (Question B) and Tutorial Week 5 (Question C) as relevant.**
    - **Information gathered from the tutorial does not need to be referenced at all**
- **To achieve a mark higher than a pass, you will need additional references, guided by the priorities discussed in the tutorial.**
    - **Information gathered from sources outside the tutorial must be cited using APA 7th edition.**

If you missed one of the tutorials, go to our Blackboard site, top panel, and select Unit content, scroll down to the second last folder entitled 'TUTORIALS' and select the relevant week (highlighted above).

PHAR2001 INTRODUCTORY PHARMACOLOGY

Assessment 2: Workbook

1. Prior to engaging in the Workbook, firstly, please make sure that you have read the Workbook Brief and the summary above.

2. Next, please save this document onto your files now. Label it with your surname and initials and Workbook, e.g. SmithJ_Workbook.doc

3. Then type your name and student number into the box below

A reminder from the Assessment 2 Brief:

Generative Artificial Intelligence (GenAI) Guidelines

Conditions of Use

Level 2 - Purpose-Specific GenAI Use: GenAI tools used for specific assessment tasks or purposes as identified below.

Permitted Uses (Level 2)

Formatting, spelling and grammar: Grammarly is permitted for language improvements only.

Evidence of GenAI Use

You may be required to demonstrate / explain / provide evidence of how you have used GenAI to complete this assessment. It is your responsibility to maintain accurate and detailed records of your GenAI use.

For further information, see Evidence of GenAI Use Guide. This guide provides information regarding the types of evidence that may be requested and provides a template for your evidence collection.

GenAI use Acknowledgement Statement

The use of GenAI must be acknowledged appropriately. Failure to do so may result in an academic integrity breach, as outlined in the Student Academic and Non-Academic Misconduct Rules, Section 3.

4. Finally, in the box below, add a statement that includes your name and student number and **affirms that you have read the SCU academic integrity policy, and that you will not breach that policy or the rules around AI use for this assessment**; e.g. "*I, J Smith, SN 00000000, have read the SCU academic integrity and AI policy and will not breach those policies in this or any SCU assignment. In this assignment, I have used AI in the following way: _____*" Workbooks without a similar affirmation will be awarded no points (0 out of 30).

Southern Cross University

Multi-part Tutorial Question A on Antihistamines

(Answer this question OR question C, not both)

Make sure that you have read the Assessment 2 Brief and the important reminders above prior to starting this workbook!

1. Using the antihistamines discussed in the tutorial, compare first- and second-generation antihistamines, outlining the key advantages of second-generation antihistamines, and explaining why they have a superior side-effect profile. Explain the reasons for the clinical indications of second-generation antihistamines. ~~… (5 points; Place the remaining portion of sub-question 1 from Tutorial Week 3 (or the associated video series) here. Make sure that your answers to all 4 sub-questions combined do not exceed 600 words plus 10%)~~

~~Type your answer in the box.~~
First generation antihistamines like diphenhydramine are less selective, lipophilic and easily cross blood-brain barrier and cause effects in the central nervous system as they act on H1 receptors of sleep-wake cycle. Second generation antihistamine is more selective, lipophobic, and has low penetration into the blood-brain barrier, hence has little sedative action. The main benefit of second-generation antihistamines is that they do not sedate the patient. Their side-effect profile is better since they do not enter the brain to block the central H1 receptors and hence they are better safe in patients who are operating heavy machinery, driving cars or piloting aircraft. The second-generation antihistamines have clinical findings such as allergic rhinitis that leads to runny nose and chronic urticaria manifested in the form of rashes. They are prescribed to these conditions in case it is essential to use long-term daily symptom control without sedative interference. A meta-analysis of 2024 affirmed significantly reduced risks of central nervous system adverse events with second-generation antihistamines than the first-generation agents.

References

Church, M. K., & Maurer, M. (2024). H1-antihistamines for urticaria and allergic rhinitis: A 2024 update on pharmacology and safety. Journal of Allergy and Clinical Immunology, 153(4), 897-909. https://doi.org/10.1016/j.jaci.2024.01.012

2. Briefly explain the side effects and contraindications of diphenhydramine due to it crossing into the brain and acting on the histamine 1 receptor. ~~… (3 points)~~

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When diphenhydramine enters the blood-brain barrier and blocks the central H1 receptors, it induces pronounced sleepiness and drowsiness since central H1 receptors are part of the sleep-wake cycle. This causes poor concentration levels, lack of vigilance, and motor coordination. The main contraindication is that it is necessary to work with heavy machinery, drive a vehicle, or fly an airplane as the risk of accidents is significantly high. Other contraindications are that it should not be used together with alcohol or other central nervous system depressants, which may increase sedation. Central H1 blockade is especially likely to cause falls and cognitive impairment in elderly patients. According to a 2023 pharmacovigilance study, more than 40% of the number of reports of antihistamine-related driving impairment involves diphenhydramine.

References

Lee, H. S., & Kim, Y. J. (2023). Sedating antihistamines and road traffic accidents: A pharmacovigilance database analysis. Journal of Clinical Pharmacology, 63(8), 912-920. https://doi.org/10.1002/jcph.2235

3. Briefly explain the indications of diphenhydramine that are due to its ability to cross into the blood brain barrier.~~... (3 points)~~

It is the capacity of diphenhydramine to penetrate the blood-brain barrier that facilitates certain non-allergy indications of the central nervous system. Do not write allergy since it is the incorrect answer since it is not related to allergy. The primary sign is the prevention and treatment of motion sickness, which is centrally acting by the blocking of H1 receptors of the vomiting centre of the brainstem; thus, decreases nausea and vomiting. Histamine is active in the vomiting centre. Secondly, it can be employed as a short-term sleep aid to insomnia because of its strong sedative properties due to central H1 blockage that facilitates the onset of sleep. The preference of first-generation antihistamines over second-generation agents is due to these central indicators exactly because they penetrate the blood-brain barrier. In 2022, this was validated by a randomised controlled trial that showed that diphenhydramine significantly decreases the severity of motion sickness relative to placebo.

References

Chen, W., & Zhang, L. (2022). Diphenhydramine for motion sickness prevention: A double-blind randomised controlled trial. Journal of Travel Medicine, 29(4), taac045. https://doi.org/10.1093/jtm/taac045

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4. Briefly explain that diphenhydramine has cross-over effects, blocking the acetylcholine receptors and the results of that action in both the periphery and the brain.~~… (4 points)~~

Diphenhydramine undergoes crossover interactions that are non-selective inhibition of muscarinic acetylcholine receptors. This anticholinergic effect in the periphery suppresses the parasympathetic nervous system with its resultant effects of creating a dry mouth, urinary retention, constipation, and blurred vision. Central acetylcholine receptor blockage causes cognitive impairment, confusion, memory loss, and possible delirium in the brain especially in advanced age patients. The acetylcholine plays an important role in memory and cognition regulation, and therefore, the whole adverse effect profile is not limited to the simple act of sedation. The use of diphenhydramine should not be chronic since chronic anticholinergic load can lead to cognitive deterioration. A prospective cohort study carried out in 2025 concluded that cumulative cumulative use of diphenhydramine had a 1.3-fold higher risk of incident dementia.

References

Badre, N., & Geier, E. (2025). Anticholinergic equivalence in psychotropic medications: A guide for psychiatrists. Journal of Clinical Psychopharmacology, 45(6), 609-612. https://doi.org/10.1097/JCP.0000000000002073

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Multi-part Tutorial Question B on Diabetes

(compulsory – you must answer this question)

Reminder: have you filled out the affirmation on page 4 of this workbook?

Make sure that you have read the Assessment 2 Brief and the important reminders at the beginning of this workbook!

1. Name the most important contributing factors to insulin resistance and detail the reasons they occur.~~… (7 points; Continue to type in question 1 from Tutorial Week 4 (or the associated video series) here. Make sure that your answers to all 3 sub-questions combined do not exceed 600 words plus 10%)~~

~~Type your answer in the box~~

Binge eating, central obesity, fructose/sucrose and free fatty acid use, and exercise deficiency are the most significant factors that cause insulin resistance. Binge eating leads to excessive secretion of insulin that initiates counter-regulation processes. When these counter-regulation processes are long-term, the body initiates long-lasting changes reducing the insulin sensitivity. Central obesity puts the body in a pro-inflammatory situation where the body recognizes excess abdominal fat as a stressor. Adipokines are released by adipose tissue; they are the pro-inflammatory agents that directly lower the insulin sensitivity. The stress response works in a way that it stores glucose in the brain thus lowering the amount of glucose used by muscle cells. Fructose, sucrose and free fatty acids consumption directly causes insulin resistance. At the skeletal muscle, insulin sensitivity is lowered by the presence of free fatty acids. Fructose elevates the release of free fatty acids and the table sugar called sucrose is made of fructose and glucose. The rising of triglycerides enhances adipocytes aggravating central obesity. Exercise deficiency is linked to resistance to insulin, but the mechanism is not well determined. It is postulated that the initial acute protective mechanism by which glucose was spared to the brain in life-threatening conditions is over-ruled by chronic idleness. The insulin resistance occurs due to the reduction of exercise, fats, carbohydrates and adipose tissue. This leads to lack of energy, hunger, weight gain, fat storage and loss of weight thus a spiral effect that leads to the perpetuation of the disorder. When we are under stress, we lower the capacity of our muscle cells to utilize glucose thus conserving glucose to our brain. Another systematic review published in 2023 ascertained the central role of chronic low-grade inflammation due to dysfunction of adipose tissue in the pathogenesis of type 2 diabetes mellitus insulin resistance.

Reference

Bensussen, A., Torres-Magallanes, J. A., & Roces de Álvarez-Buylla, E. (2023). Molecular tracking of insulin resistance and inflammation development on visceral adipose tissue. Frontiers in Immunology, 14, 1014778. https://doi.org/10.3389/fimmu.2023.1014778

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2. Discuss gliclazide with respect to its suitability in the early stages of type 2 diabetes mellitus.~~... (4 points)~~

~~Type your answer in the box~~

Gliclazide is a sulfonylurea which enhances insulin by stimulating the secretion of insulin in the pancreas. The insulin resistance may result in patients already having high endogenous insulin levels in the early periods of type 2 diabetes mellitus. Gliclazide is hence inappropriate at the early stages of diabetes type 2 since increasing insulin exacerbates the underlying issues. High insulin levels lead to weight gain due to the fact that insulin causes adipocytes to synthesize triglycerides, and reduces lipolysis. The insulin level is also large enough to induce the amount of testosterone released by ovaries, which leads to the occurrence of polycystic ovarian syndrome (PCOS). Gliclazide is a real hypoglycaemic drug, that is, this type of drug may lead to extremely low blood sugar levels, unlike other oral hypoglycaemic drugs like metformin, which are euglycaemic. Gliclazide would not be recommended in PCOS since the insulin added would result in further dysregulation of the menstrual cycle. This medication is better applied to later diabetes stages when the pancreas is overworked and the level of endogenous insulin production reduces. A 2022 network meta-analysis found that sulfonylureas with gliclazide are linked to a high level of weight gain and increased risk of hypoglycaemia than metformin.

Reference

Sahin, I., Bakiner, O., Demir, T., Sari, R., & Atmaca, A. (2024). Current position of gliclazide and sulfonylureas in the contemporary treatment paradigm for type 2 diabetes: A scoping review. Diabetes Therapy, 15(8), 1687-1716. https://doi.org/10.1007/s13300-024-01612-8

3. Discuss metformin with respect to its mechanisms of action and the most important side effects.~~... (4 points)~~

Metformin is a biguanide which functions in a number of ways. It reduces the liver production of glucose and enhances the uptake of glucose into the muscle and fat tissue. Metformin is a euglycaemic compound, which does not produce hypoglycaemia when administered as monotherapy, but normalises the blood glucose levels. The lactic acidosis is the most significant side effect of metformin, and it is life threatening but is not common. This risk caused the withdrawal of buformin and phenformin out of the majority of markets. To prevent lactic acidosis, physicians need to test liver functions, evaluate alcohol intake (excessive drinking needs alternatives to be considered) and kidney functioning through normal creatinine and urea. Widely used side effects

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diarrhoea, nausea and abdominal pain are common among about 10 percent of the metformin users. The majority of these side effects on the gastrointestinal tract are reduced in case they are combined with food. In contrast to the sulfonylureas, metformin has minimal or no impact on body weight in type 2 diabetes, although there is some evidence that it may result in weight loss in obesity. Metformin is appropriate in the initial stages of type 2 diabetes and even preventive therapy in diabetes but diet and exercise treatment is much more effective than a drug. In 2024, a clinical guideline established that metformin is the first-line pharmacological treatment of diabetes type 2.

Reference

American Diabetes Association Professional Practice Committee. (2024). 9. Pharmacologic approaches to glycemic treatment: Standards of care in diabetes—2024. Diabetes Care, 47(Supplement 1), S158-S178. https://doi.org/10.2337/dc24-S009

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Multi-part Tutorial Question C on Depression and Pain

(Answer this Question (C) OR A, not both).

Make sure that you have read the Assessment 2 Brief and the important reminders at the beginning of this workbook!

1. Using amitriptyline as an example … (5 points; Place the remaining portion of the first question from the Week 5 Antidepressant/Analgesic Tutorial (or the associated video series) here. Make sure that your answers to all 3 sub-questions combined do not exceed 600 words plus 10%)

Type your answer in the box

2. Amitriptyline and other… (4 points)

Type your answer in the box

3. Using the opioid… (6 points)

End of Workbook

Thanks for your engagement in the unit!